



International University of Business
Agriculture and Technology



DEVELOPING THE NETWORK SYSTEM AND LOAD BALANCING FOR RADIAL INTERNATIONAL LTD.

Prepared for
Practicum Board of IUBAT

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INTRODUCTION

Specific objectives

- Redesigning the Network System.
- Dynamic IP allocation.
- Redundancy implementation.
- Load Balancing.

COMPANY OVERVIEW

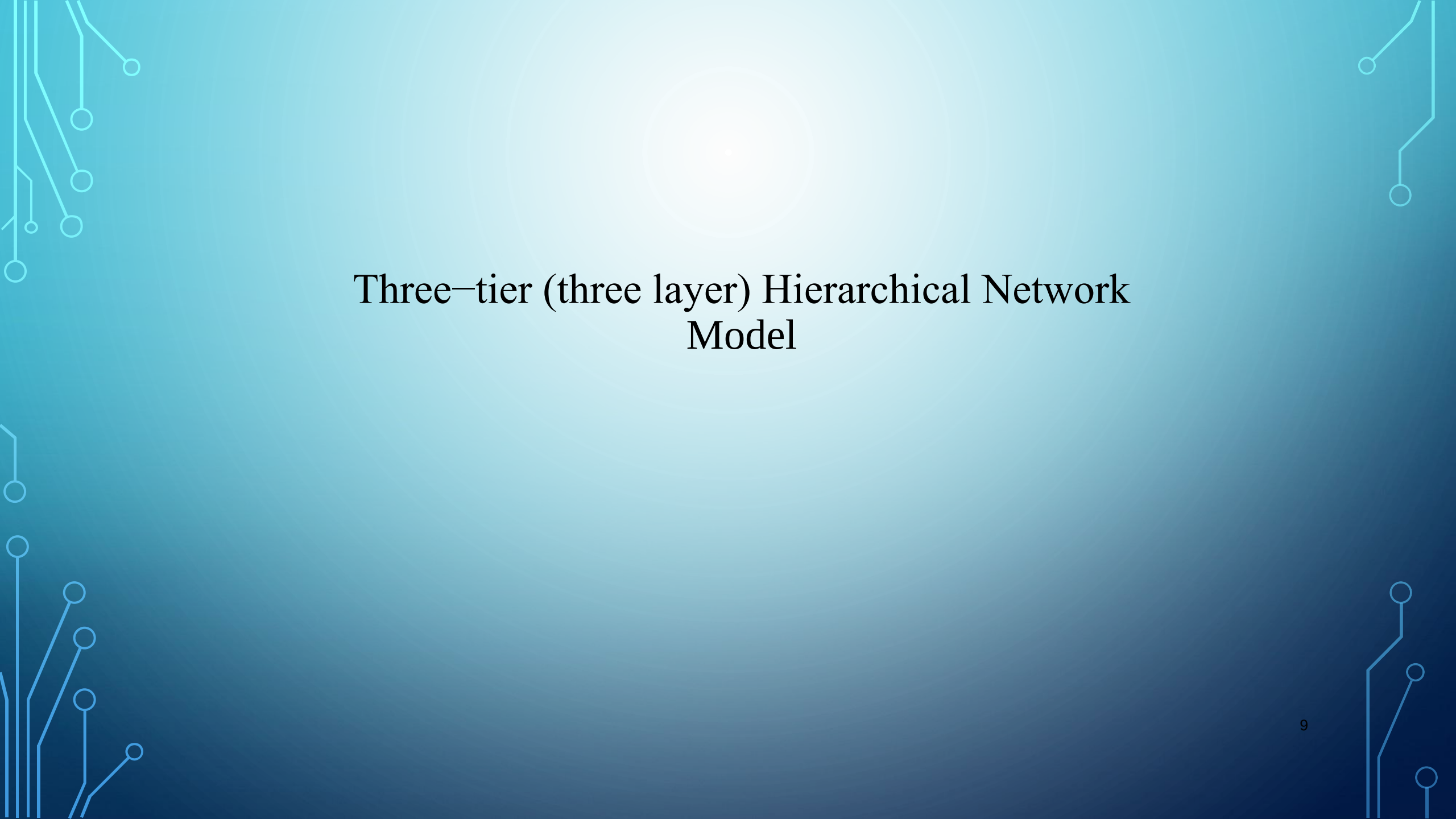
RADIAL INTERNATIONAL LTD.

- A Garment Manufacturer Company.
- Established in 1991.
- Composite knitwear company with knitting, dyeing, cutting, sewing and finishing.
- Oekotex certified company.

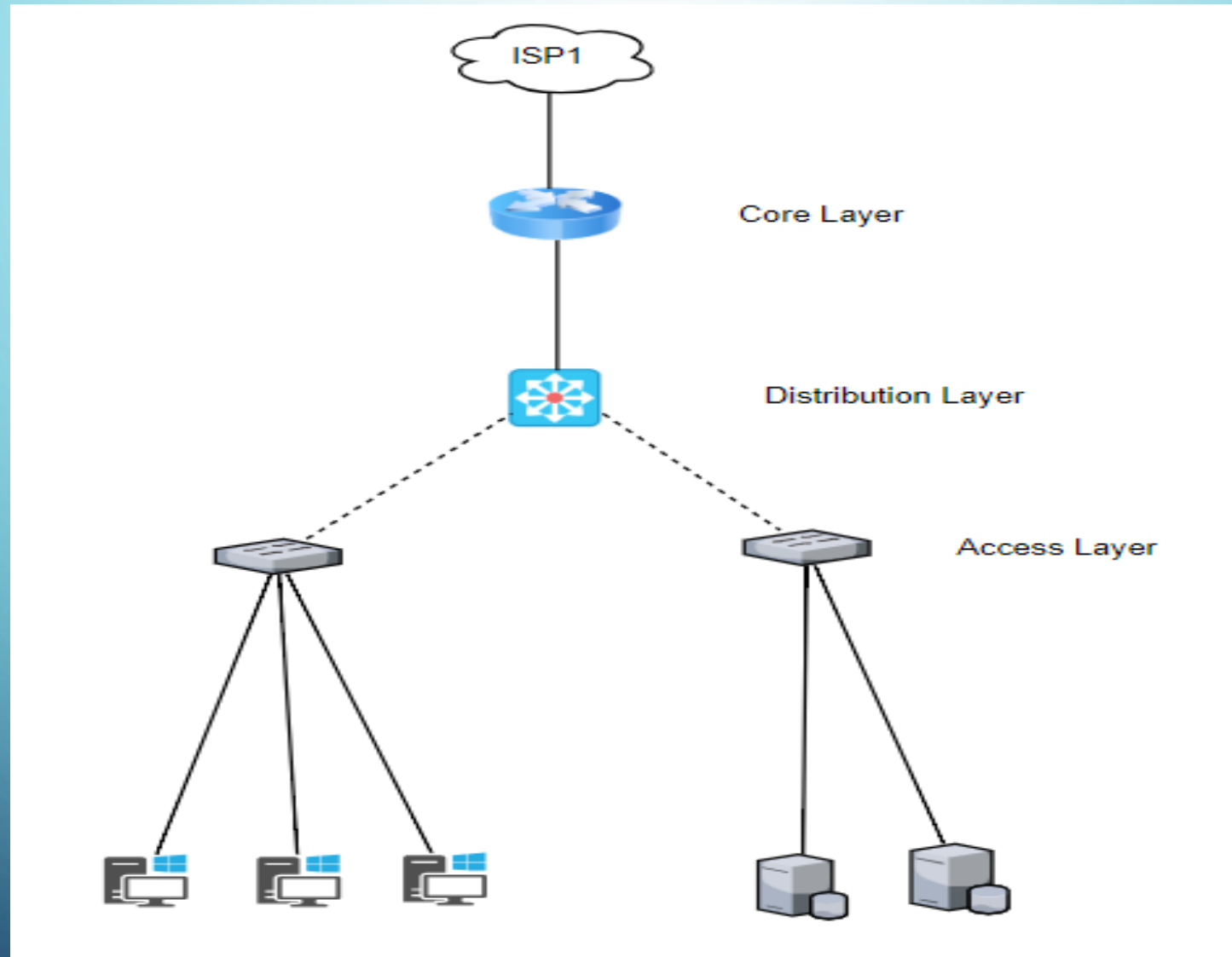
EXISTING NETWORK

Branch Offices:

- Gazipur Branch Office (Concern Office).
- Baridhara Main Office.

The background features a blue gradient with a central glowing circle and faint concentric circles. Decorative circuit-like lines with nodes are positioned in the corners.

Three-tier (three layer) Hierarchical Network Model





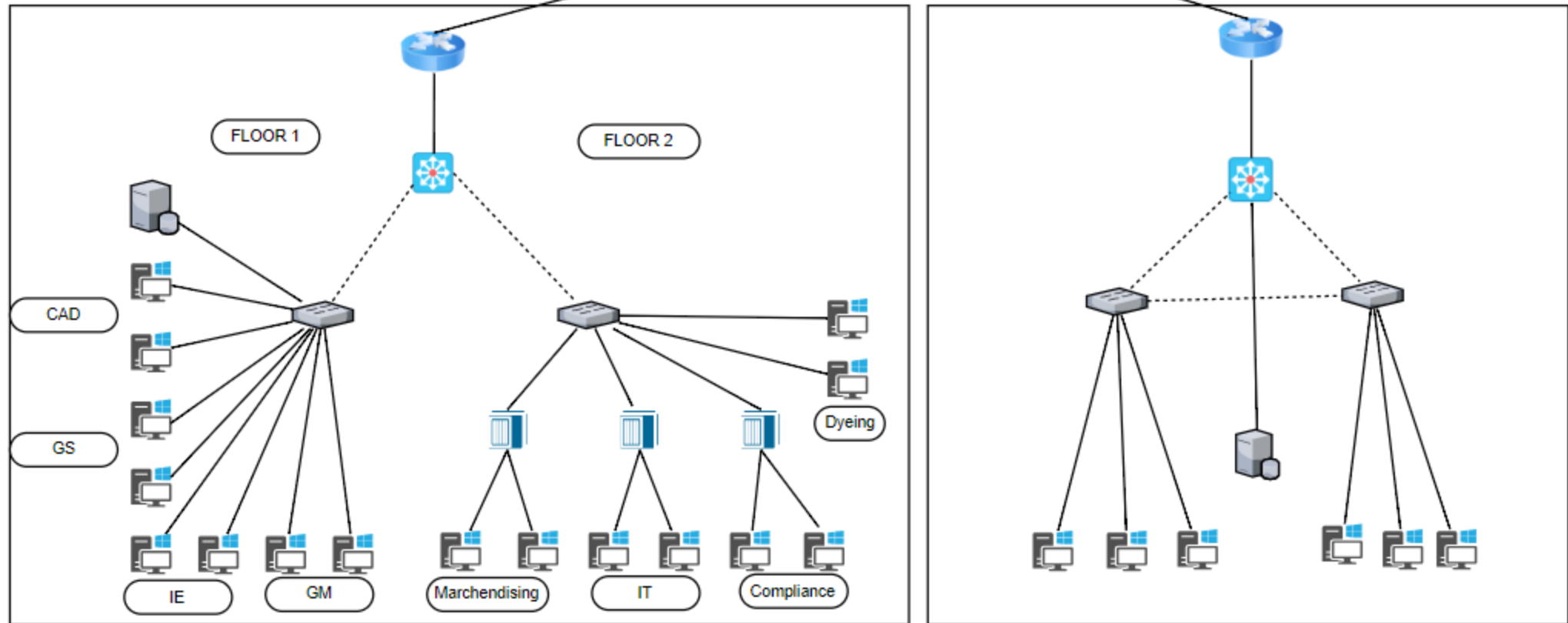
Gazipur Branch Office



ISP



Baridhara Head Office



Departments in Existing System

- I. General Manager (GM).
- II. CAD Department.
- III. Garments Store (GS) Department.
- IV. Industrial Engineering Department (IE).
- V. Information Technology (IT) Department.
- VI. Dyeing Department
- VII. Marchendising Department.
- VIII. Compliance Department.

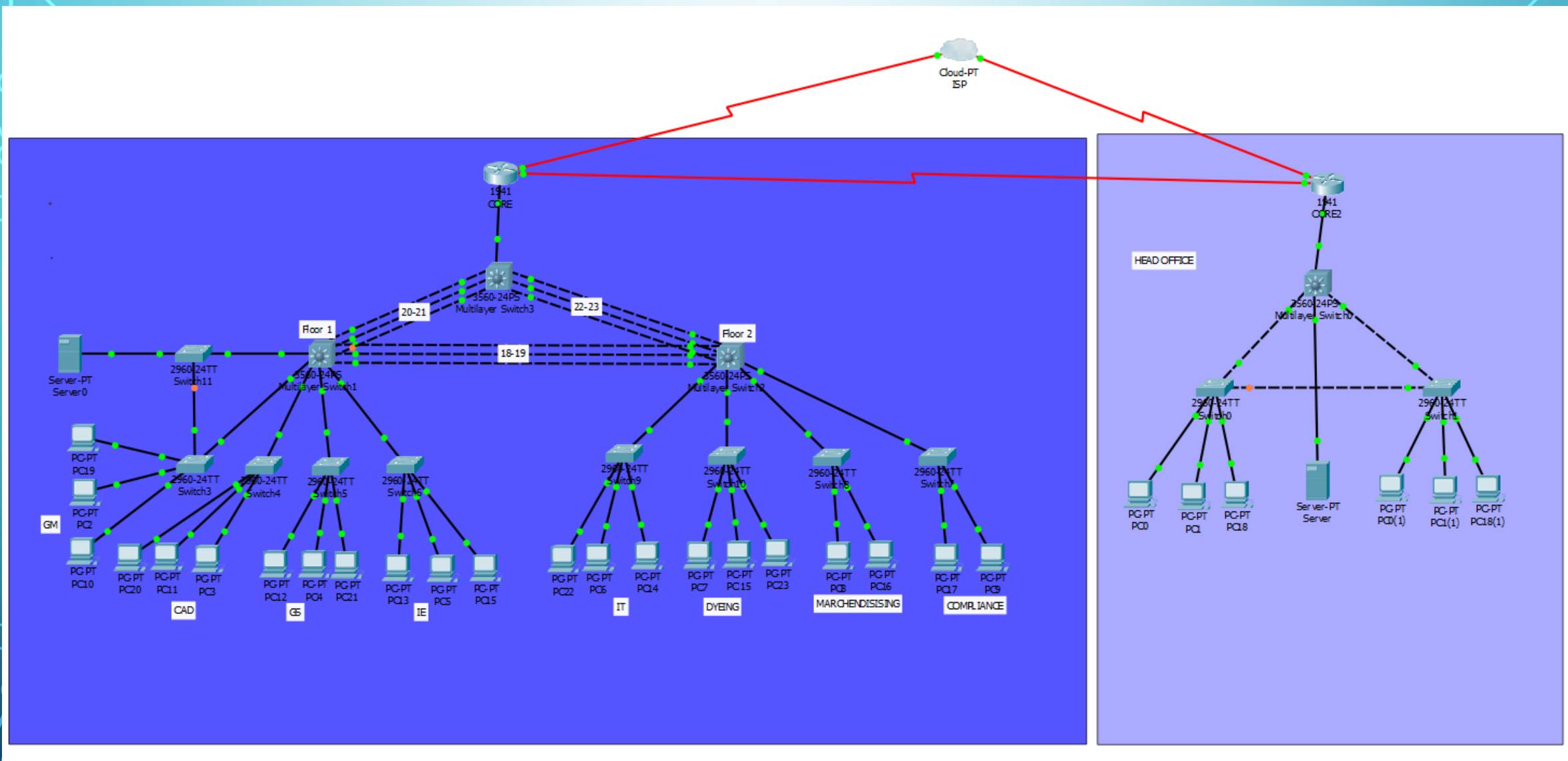
Communication Structure between Departments

Leased Line

Limitation of Existing Network

- Use of old devices
- No backup path for instant solution.
- No load balancing method.
- No separation of different departments.
- No dynamic IP allocation system.

PROPOSED NETWORK



Redesigning the Network System.

- Replacing the network device HUB with Cisco catalyst switches.
- Separating the departments from each other.
- Providing switches for individual department.
- Controlling access.
- Configuration: VLAN, Inter VLAN, Access Control List.

Dynamic IP Allocation.

- Separate IP series for different departments.
- Automatic Private IP Addressing (APIPA) feature.
- Automatic IP configuration.
- Dynamic IP allocation.
- Configuration: Dynamic Host Configuration Protocol.

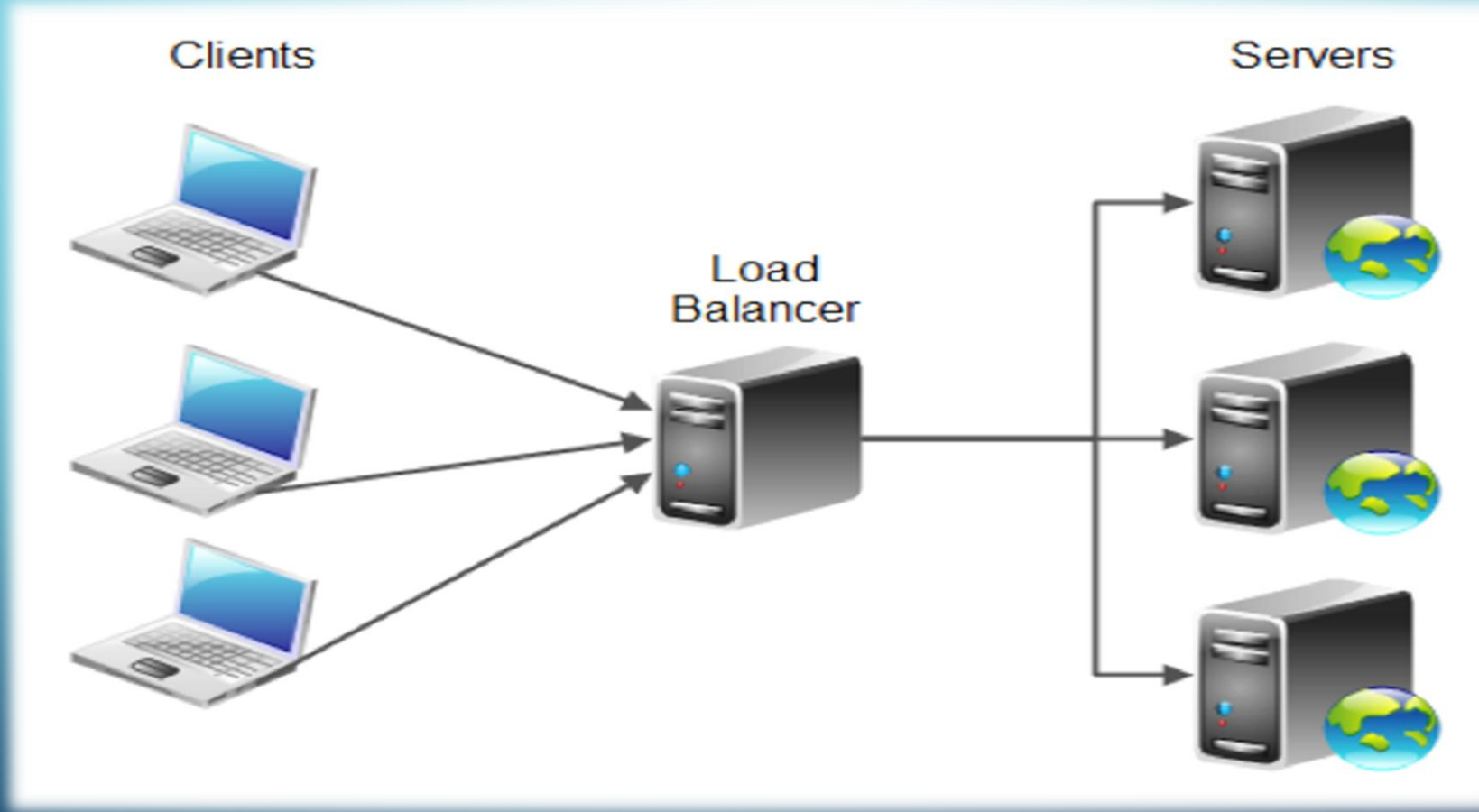
Redundancy Implementation.

- In place of one multi-switch total three multi-switches are used.
- Multiple paths are configured as a single group channel.
- Providing backup path for the server.
- Configuration: STP (Spanning tree protocol), Ether-Channel.

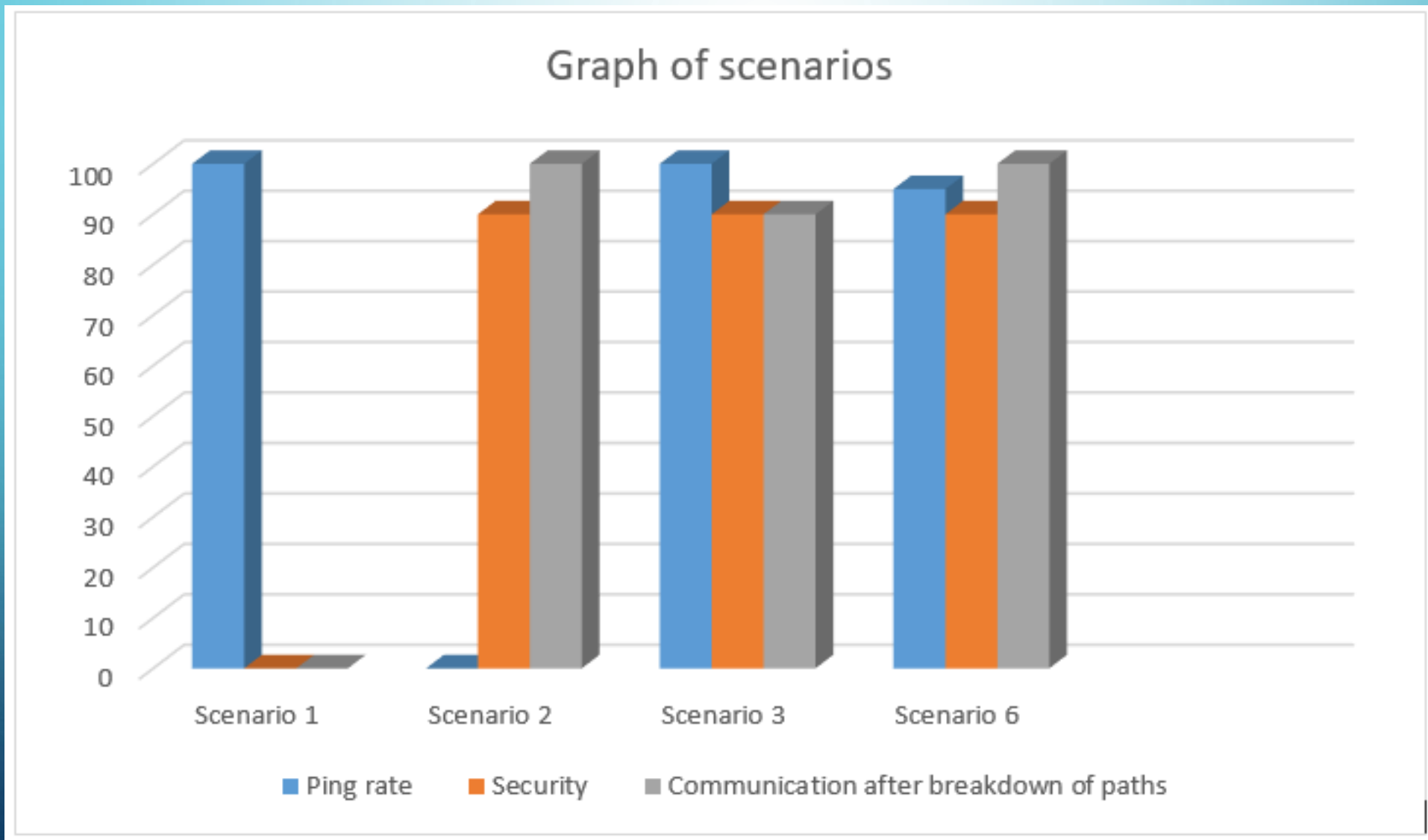
Load Balancing

- By providing multiple links between multilayer switches the data transmission load between department has been balanced.
- Fast data transmission
- Configuration: Ether-Channel

Load Balancing Features with SNAT (Secure NAT) with the Help of Load Balancer



COMPARISON



Significance of Proposed Network

- Dividing the departments by redesigning the network system.
- Controlling the access permission.
- Providing dynamic IP configuration.
- Providing redundant path to make the communication faster.
- Load balancing of data transmission between the departments.

DEMONSTRATION

CONCLUSION



Thank
You